CertJudge: Evaluating Lean Formal-Code With Falsifiable Properties

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Validating LLM judges one human study at a time does not scale

- AI agents now write Lean 4 code **and** the theorems/specs meant to verify it — far faster than humans can inspect.
- The Lean kernel checks *proofs*, not whether the *right theorem* was proposed: a candidate can compile yet be under-specified, vacuous, or semantically shifted.
- So an LLM judge scores each candidate against a gold reference — but every new judge, prompt, model, or threshold demands a **fresh human-agreement study**.

Can one human calibration be amortized across future judges, via checks that are automatic and falsifiable?

Key idea: certify the judge — don't re-label it

- **CertJudge**: humans rate a small panel **once** (n=75 candidate–gold Lean pairs); every new judge configuration is then screened by **automatic, falsifiable property checks**.
- Controlled perturbations test four behaviors, aggregated into a **variance-aware trust index** $\mathrm{TI}_{\mathrm{var}}$.
- Judges with higher $\mathrm{TI}_{\mathrm{var}}$ agree more with humans — certification predicts *which judge to trust*, with no new labels.

Four falsifiable properties → one trust index

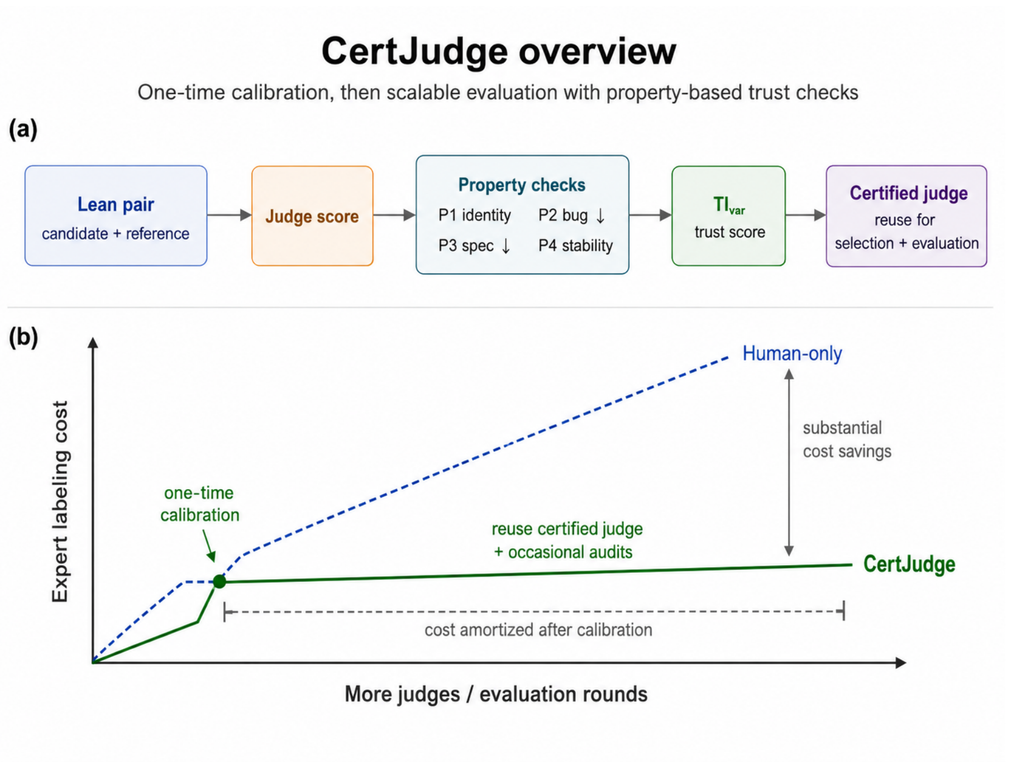


Figure 1. Label once, then certify automatically: property checks plus $\mathrm{TI}_{\mathrm{var}}$ replace per-judge human studies, so labeling cost stops growing with the number of judges.

- **P1 Identity:** $s(X, X) = 1$ — an artifact scored against itself gets the maximum.
- **P2 Bug monotonicity:** score falls as semantic bugs are injected.
- **P3 Spec monotonicity:** score falls as theorems/tests are removed.
- **P4 Stability:** score invariant across repeats and semantics-preserving rewrites.

$$\mathrm{TI}_{\mathrm{var}} = \exp\left(\tfrac{1}{3}\log C_f + \tfrac{1}{3}\log(1 - \widetilde{\mathrm{VarCorr}}) + \tfrac{1}{3}\log(1 - \widetilde{\mathrm{VarScore}})\right)$$

C_f: geometric aggregate of P1–P4; the variance terms penalize unstable judges. Multiplicative: **one failed property collapses the index**.

Calibration setup

VeriBench Lean 4 artifacts [1]: n=75 aligned candidate–gold pairs from 24 problems (3 agent generations per problem), two human passes each on a 0–5 rubric. 8 judge configurations per family: GPT-5.3-Codex, Claude Opus 4.7, Qwen3-Coder.

Result: $\mathrm{TI}_{\mathrm{var}}$ predicts judge-level human alignment

$$\rho = 0.833$$

validation-set Spearman between $\mathrm{TI}_{\mathrm{var}}$ and human agreement (avg labels; 0.905 pass1, 0.738 pass2)

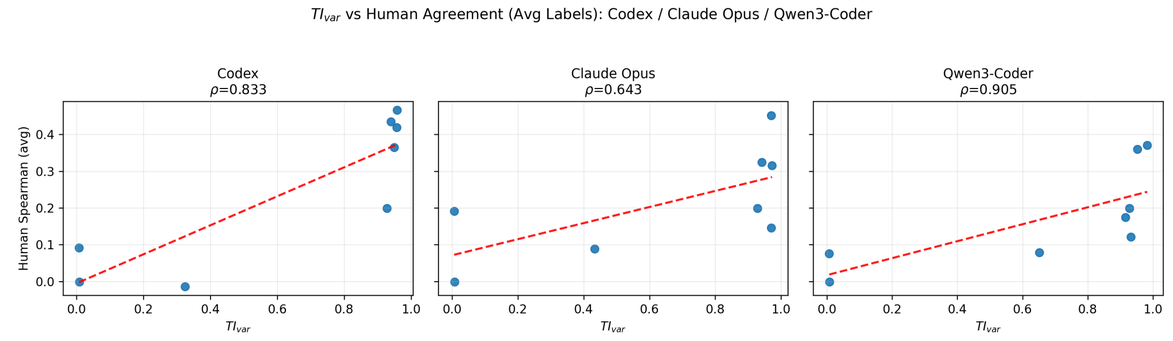


Figure 2. Each point is one judge configuration: higher $\mathrm{TI}_{\mathrm{var}}$ → higher human agreement, in **all three judge families** (avg labels).

Catches what agreement alone misses: an intentionally broken judge (Bad-Stringy) still reaches ≈0.45 human Spearman, yet fails weak monotonicity (12.5%) — $\mathrm{TI}_{\mathrm{var}}$ flags it.

Stress tests keep the judge honest

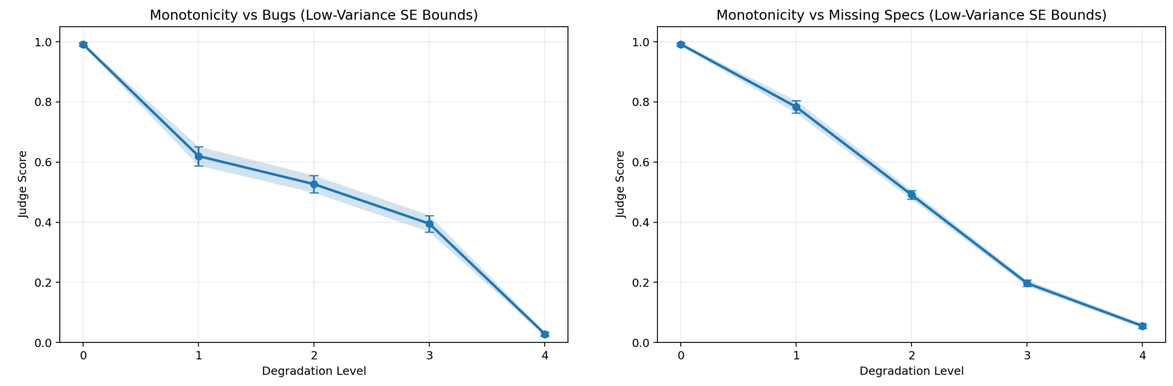


Figure 3. GPT-5.3-Codex judge, 184 VeriBench files: scores fall as bugs are injected (left, $\rho = -0.808$) and as specs are deleted (right, $\rho = -0.899$).

Known failure mode: aggressive α-renaming scores 0.399 vs. ≥ 0.849 for milder semantics-preserving rewrites — LLM judges key on surface strings the Lean kernel ignores.

The certified judge ranks theorem-generation agents

On VeriBench, the calibrated judge ranks feedback-driven agents above plain prompting (avg normalized score): **DSPy ReAct 0.615** > Trace++ 0.588 > Trace+ 0.477 > **baseline 0.470** — a downstream sanity check, not independent proof of correctness.

Takeaways

- **Amortized oversight:** one n=75 human panel + automated falsifiable checks screen every new judge configuration — no re-labeling.
- **Measure trust, don't assume it:** behavioral properties expose failures (broken baselines, α-renaming brittleness) that human-agreement scores miss.
- **Scope:** *validation-set calibration evidence, not universal reliability* — a locked held-out test on an independent pool (5 raters, 343 labels, 78 pairs) is next.

References

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Code